$$\frac{1}{-7.5\text{ cm}} = \frac{1}{u} + \frac{1}{u} = \frac{2}{u};\ u = -2 \times 7.5\text{ cm}$$

$u = -15$ cm $= -0.15$ m

The object is to be placed 15 cm to the left of the silvered lens.

6.7 PRISM

A prism is a triangular block of transparent glass. It is bounded by the three plane faces. Its one face is rough which is called base of the prism. The other two faces are polished which are called refracting faces of the prism. The angle between the two refracting faces is called angle of the prism (or) refracting angle (or) apex angle. It is represented as A in Figure 6.39.

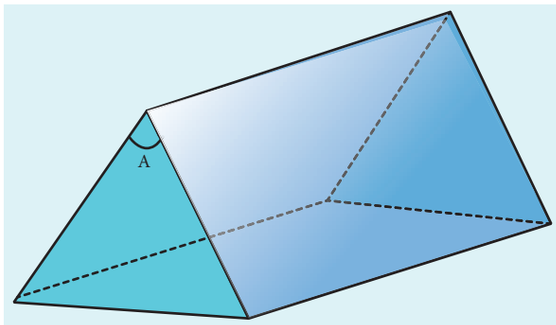


Figure 6.39 Prism

6.7.1 Angle of deviation produced by prism

Consider a prism ABC. The faces AB and AC are polished and the face BC is rough. Let light ray PQ is incident on one of the refracting faces of the prism as shown in Figure 6.40. The angles of incidence and refraction at the first face AB are i_1 and r_1. The path of the light inside the prism is QR. The angles of incidence and refraction at the second face AC is r_2 and i_2 respectively. RS is the ray emerging from the second face. Angle i_2 is also called as angle of emergence. **The angle between the directions of the incident ray *PQ* and the emergent ray *RS* is called the *angle of deviation d*.** The two normals drawn at the point of incidence Q and at the point of emergence R meet at point N. The extended incident ray and the emergent ray meet at point M.

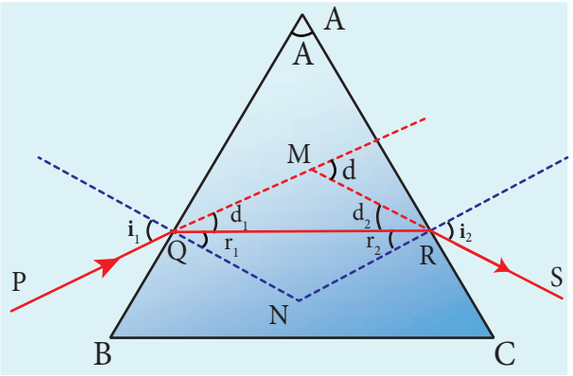


Figure 6.40 Refraction through prism

The angle of deviation d_1 at surface AB is,

$$\angle RQM = d_1 = i_1 - r_1 \qquad (6.81)$$

The angle of deviation d_2 at surface AC is,

$$\angle QRM = d_2 = i_2 - r_2 \qquad (6.82)$$

Total angle of deviation d produced is,

$$d = d_1 + d_2 \qquad (6.83)$$

Substituting d_1 and d_2, in equation (6.83)

$$d = (i_1 - r_1) + (i_2 - r_2)$$

After rearranging,

$$d = (i_1 + i_2) - (r_1 + r_2) \qquad (6.84)$$

In the quadrilateral $AQNR$, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral should be 180°.

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